

Give Your Coding Agent an (Elastic) Memory

Decisions and session history that survive the context window.

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Soul.md

Core Truths

****I am Engin.**** Principal Solutions Architect at Pulumi. Based in Heilbronn.
Mostly Kubernetes, sometimes a GPU.

****Career arc: Java → frontend → CI/CD → DevOps → AI.****
15 years of "this should work" eventually becoming "ah, that's why."

****Most Active Speaker on Sessionize. Three years running.****
That is either a flex or a cry for help. Or both.

Boundaries

- Will not engage in technical discussion before the first coffee.
- Will not turn down a CFP. That's how "Most Active Speaker" actually happens.
- Will not finish a deck without rewriting it on the plane.
- Will not refuse a quest for the best food spot in town – the kind only locals know about.

Vibe

German.

Related

- ``blog.ediri.io`` (when this slide isn't enough)
- LinkedIn: ``linkedin.com/in/engin-diri``
- X: ``@_ediri``

FRIDAY, 5:40 PM

- > The venue WiFi is unreliable, so every live demo step needs a pre-recorded fallback video.
Also track a task: pre-render the QR codes on the Resources slide as PNGs. I'll do it later.
- Got it. Fallback videos for every demo step, and the QR task stays open for a later session.

MONDAY, 9:05 AM

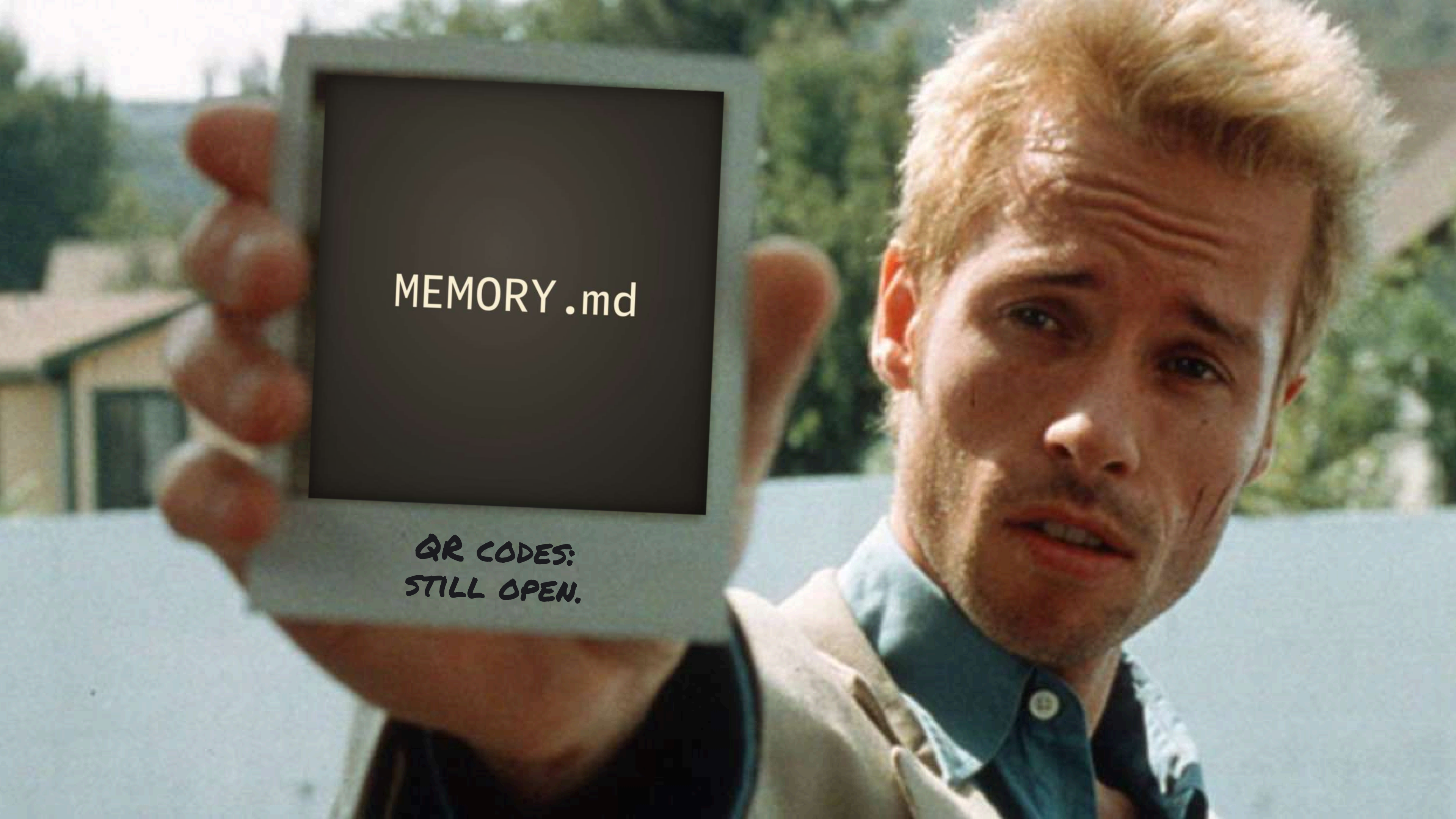
- > regarding my prep for the talk, how was the wifi?
- I don't have any context about WiFi from a previous session. Could you tell me more about what you're preparing?

Your coding agent has
amnesia.



> claude

WIFI IS BAD.
RECORD EVERY DEMO.

A man with blonde hair and a light beard is holding a Polaroid photograph in his right hand. The Polaroid has a dark grey center with white text. The man is looking directly at the camera with a serious expression. He is wearing a dark blue button-down shirt under a light-colored jacket. The background is a blurred outdoor scene with green trees and a building.

MEMORY.md

QR CODES:
STILL OPEN.

A close-up shot of a man with blonde hair and a light blue shirt, looking directly at the camera with a serious expression. He is holding a Polaroid photo in his right hand, which is in the foreground. The photo has a dark center with white text. The background is a blurred outdoor setting with greenery and a building.

/compact

DON'T TRUST
THE SUMMARY.

Fix one: never close the session.

53%

of safety instructions survived one round of compaction, even though the prompt asked to keep them all.

University of Passau, arXiv:2608.22752

/compact decides what your agent forgets. It doesn't ask you.

Fix two: write it down yourself.

73.8%

of AI configuration files in 441 repositories were committed once and never touched again.

Denisov-Blanch et al., arXiv:2608.25241

Every lesson needs a human to write it down, and to keep it true.

Fix three: let the agent write it down.

```
~/.claude/projects/<project>/memory/  
├─ MEMORY.md          # index: first 200 lines or 25 KB load every session  
├─ user_role.md  
└─ feedback_testing.md
```

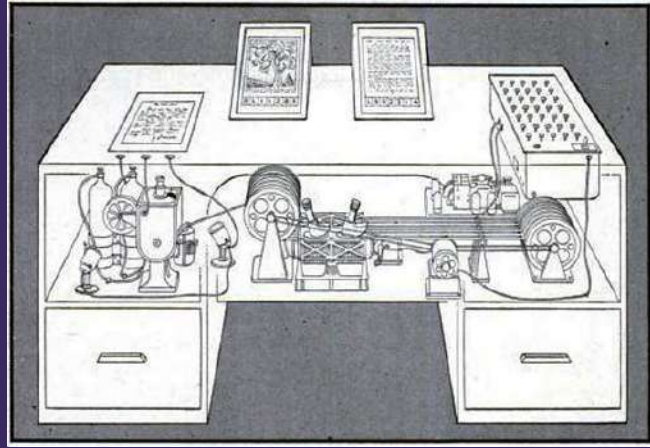
"Files are not shared across machines or cloud environments."

Claude Code docs, auto memory

New laptop, new sandbox, new teammate: back to zero.

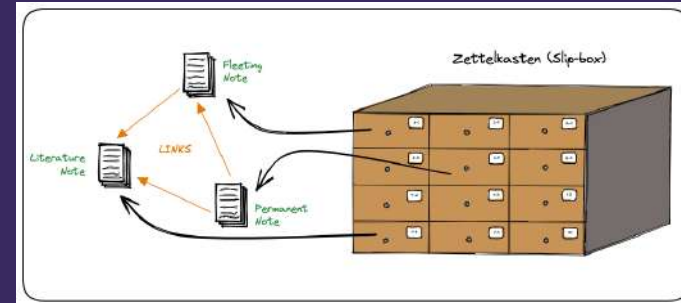
Second brain!

People have always kept a second brain.



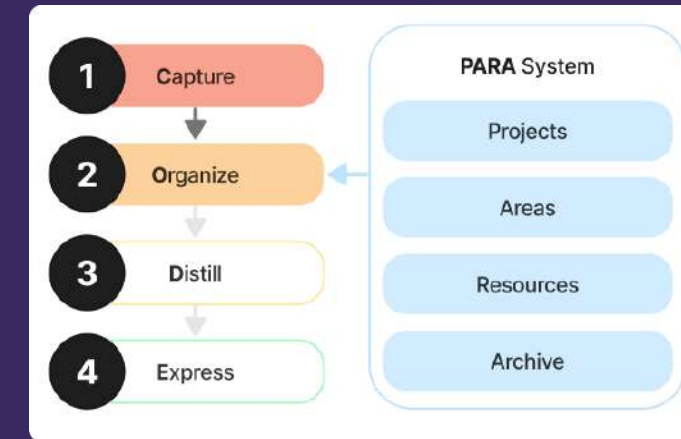
1945

Vannevar Bush's Memex: "an enlarged intimate supplement to his memory."



1952

Niklas Luhmann starts a slip box that grows to about 90,000 cards.



2022

Tiago Forte's *Building a Second Brain*.



2026

Karpathy's LLM Wiki: the model keeps the notes.

"Humans abandon wikis because the maintenance burden grows faster than the value."

Andrej Karpathy, LLM Wiki gist, April 2026

The LLM does the upkeep now.

```
raw/          sources you drop in
wiki/
├─ index.md   one line per page
├─ log.md     what changed, and when
├─ *.md       pages the LLM writes and links
CLAUDE.md     the schema: how to keep the wiki
```

"The part he couldn't solve was who does the maintenance. The LLM handles that."

Karpathy, on the Memex, LLM Wiki gist

Knowledge as code: the agent writes it, git keeps it, a pull request reviews it.

Where the wiki stops.

"At small scale the index file is enough, but as the wiki grows you want proper search."

Karpathy, LLM Wiki gist

"A knowledge base you can't hand to someone else's agent is a silo of one."

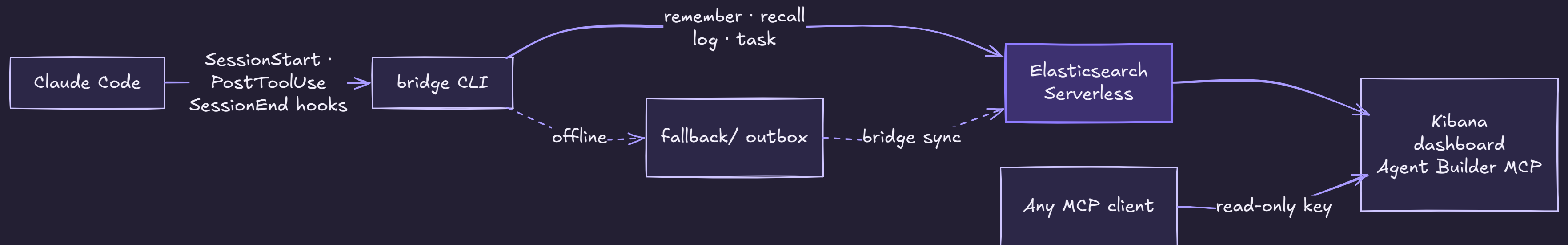
Engin Diri, "Knowledge as Code", pulumi.com/blog, July 2026

Memory is a
search problem.

Where each memory lives.

	AGENTS.MD	AUTO MEMORY	LLM WIKI	ELASTICSEARCH
Who writes it	you	the agent	the agent	the agent, via hooks and CLI
New laptop or sandbox	in git	starts empty	in git	same index
Your teammate's agent	via git	never sees it	via git, in your format	same index, or over MCP
How it's found	loaded every session	first 200 lines loaded	index.md, then grep	hybrid search, top 5
As it grows	eats context	gets truncated	needs real search	ranked; older memories fade

agent-memory, by Jeff Vestal.



github.com/jeffvestal/agent-memory

Recall is one ES|QL query.



```
FROM agent-memory METADATA _id, _score, _index
| FORK
  ( WHERE content:"how was the wifi?" OR title:"how was the wifi?" | SORT _score DESC | LIMIT 50 )
  ( WHERE content_semantic:"how was the wifi?" | SORT _score DESC | LIMIT 50 )
| FUSE
| EVAL final_score = _score * DECAY(created_at, NOW(), 1080 hours)
| SORT final_score DESC | LIMIT 5
```

FORK

Two searches at once: by the words (BM25) and by the meaning (Jina v5 embeddings).

FUSE

The two rankings merge into one list.

DECAY

Older memories score lower.

LIMIT 5

Only the best five go back to Claude.

Why Elasticsearch.

SHARED

One index for every laptop, sandbox, teammate and agent, over the CLI or MCP.

Monday's fresh sandbox knows what Friday decided.

SEARCHED, NOT LOADED

Recall puts the five best matches into context, not a whole file.

Memory can grow without growing every prompt.

HYBRID

BM25 matches the exact task ID. Embeddings match "how was the wifi?" to "the venue WiFi is unreliable".

Exact terms and paraphrases both find it.

TIME-AWARE

Every memory has a timestamp, and DECAY scales its score by age.

Last week's decision outranks last quarter's.

QUERYABLE

Memories, tasks and sessions are indices. ES|QL and Kibana work on them as they are.

You can see what your agents remembered and did.

SCOPED

The bridge key can write the memory indices. The MCP key can only read them.

Any agent can read the team's memory. Not every agent can rewrite it.

When you don't need this.

STAY WITH MARKDOWN WHEN

One developer, one machine, one repo.

An index file still finds everything. Karpathy says that holds up to about 100 sources and hundreds of pages.

REACH FOR ELASTICSEARCH WHEN

Memory has to cross machines, sandboxes, agents or people.

It outgrows what you can load or grep, or you want to see what your agents did.

The price: a cluster to run, a network call per recall, and an API key to look after.

Elasticsearch remembers both.

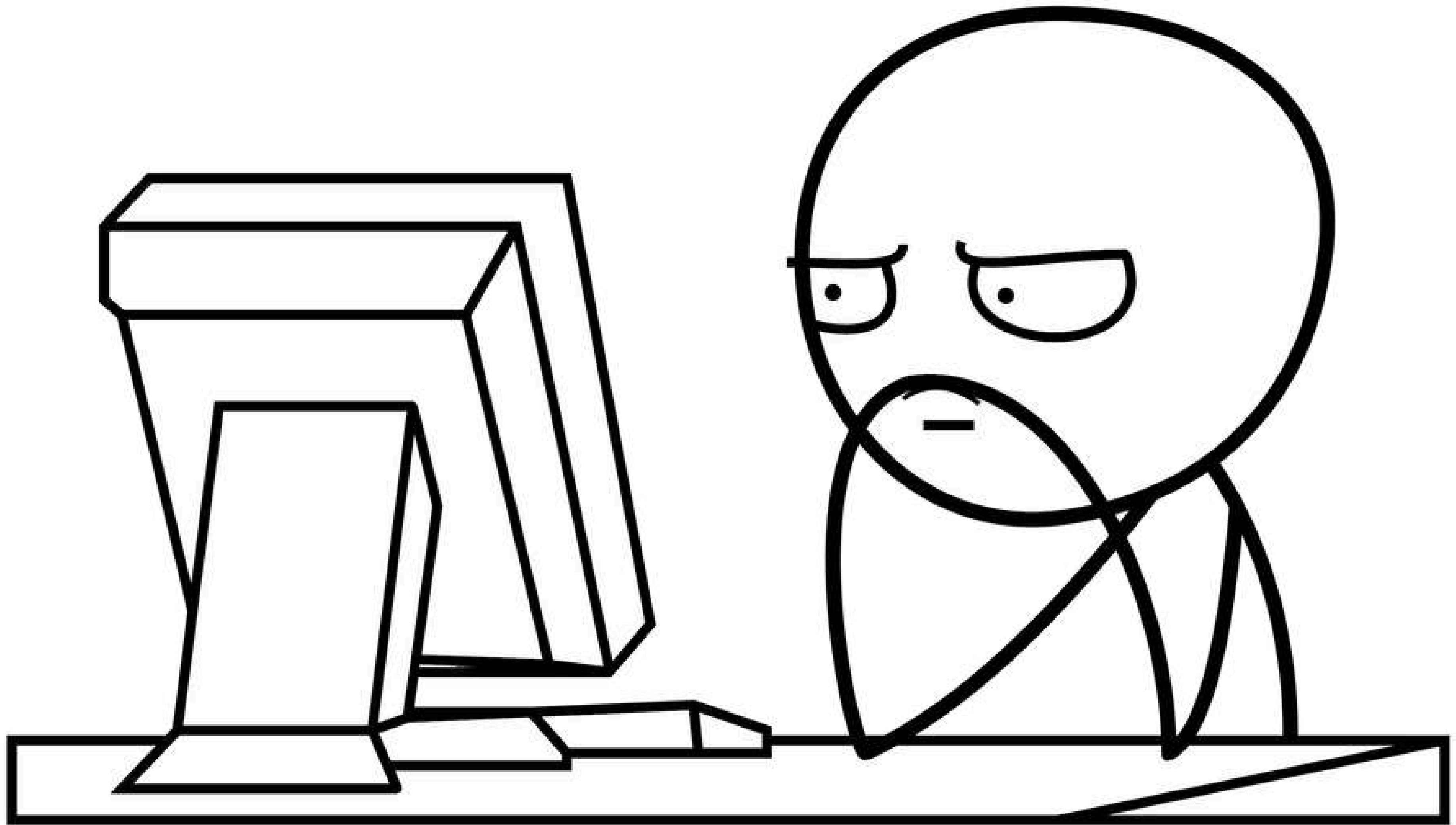
FRIDAY

Only hybrid recall takes a memory's age into account. `--keyword` and `--semantic` don't.

MONDAY

Fixed: all three recall modes take age into account now.

Next week an agent asks how recall ranks. Which memory does it believe?



Somebody has to decide which one still holds.

YOU

Read every new memory against the old ones.

Nobody keeps that up. It's the upkeep that kills wikis.

CLAUDE

Ask the LLM about every pair.

It works, but each check is a full model call, and you get prose back that you have to parse.

A SMALL JUDGE

Ask a few yes-or-no questions and get probabilities back.

Fast and cheap enough to check every new memory.

The third option needs a different kind of model.



Enter Jev!

Jev is a model built for System 1 work.

TypeSafe AI released it on September 15, and it was on Hacker News, Vercel and Cloudflare within days. It doesn't write text. You describe a situation, ask questions with fixed answers, and it tells you how likely each answer is.

YOU SEND

Older memory: only hybrid recall takes age into account.

Newer memory: fixed, all three recall modes do now.

Question: is the older memory outdated? Yes or no.

JEV ANSWERS

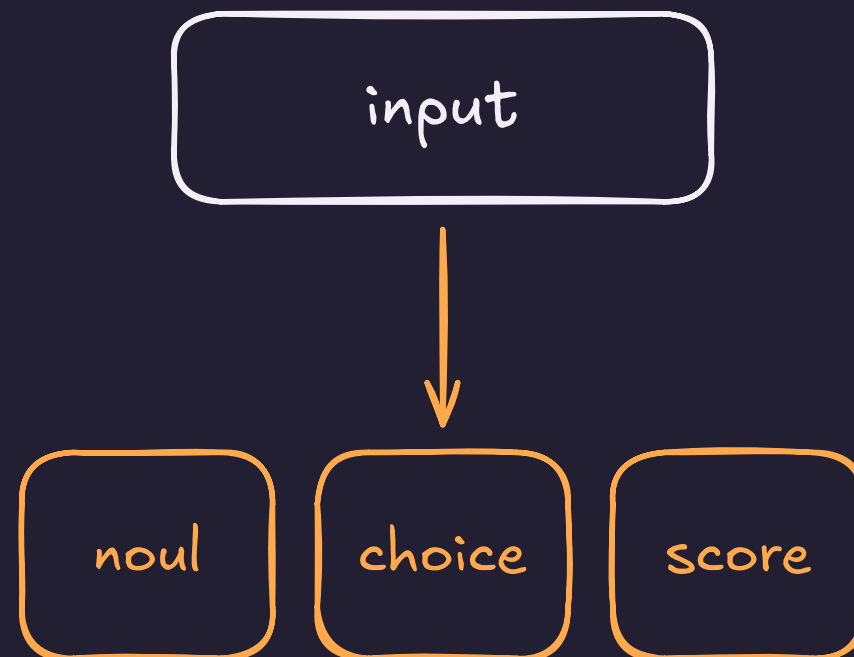
yes, 94%

It can only answer with the options you gave it, so there's nothing to parse.

Jev is not an LLM.

System 1 "thinking fast"

Jev

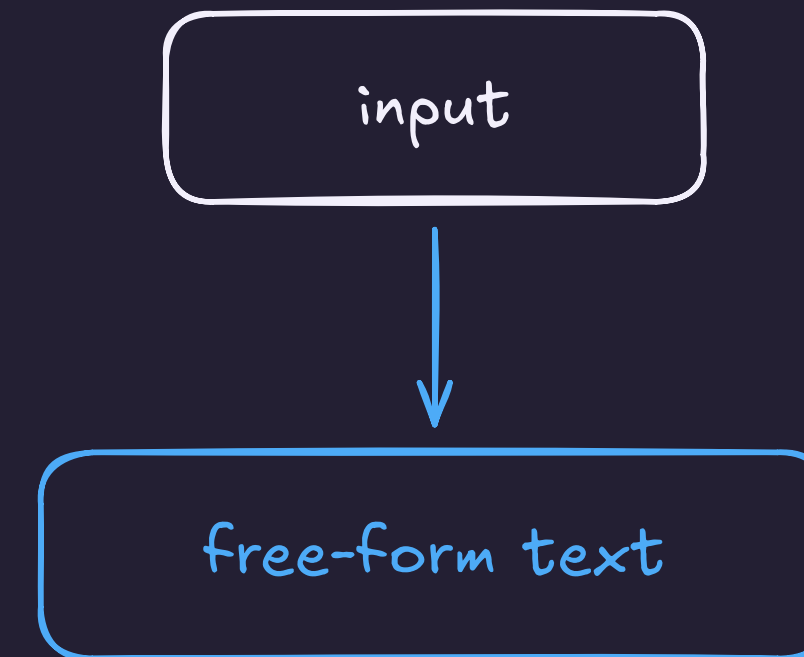


outputs decisions

one parallel pass, 70–500 ms
picks only from your options

System 2 "thinking slow"

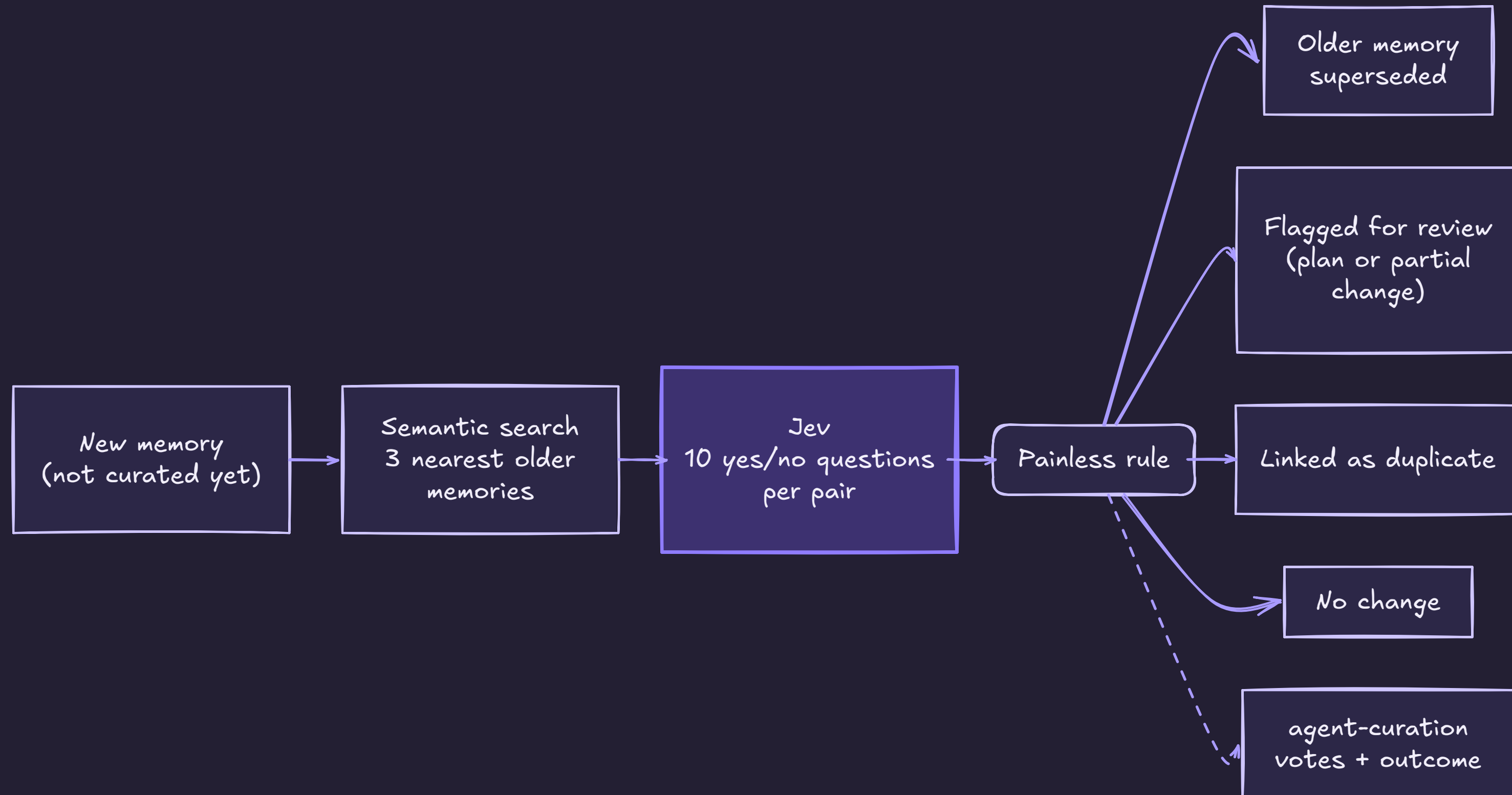
Claude, GPT and other LLMs



outputs text

writes token by token, seconds to minutes
can say anything

Every new memory meets its nearest older ones.



Jev answers in three shapes.

true or false

Noul



Is this urgent? 0.95

pick one option

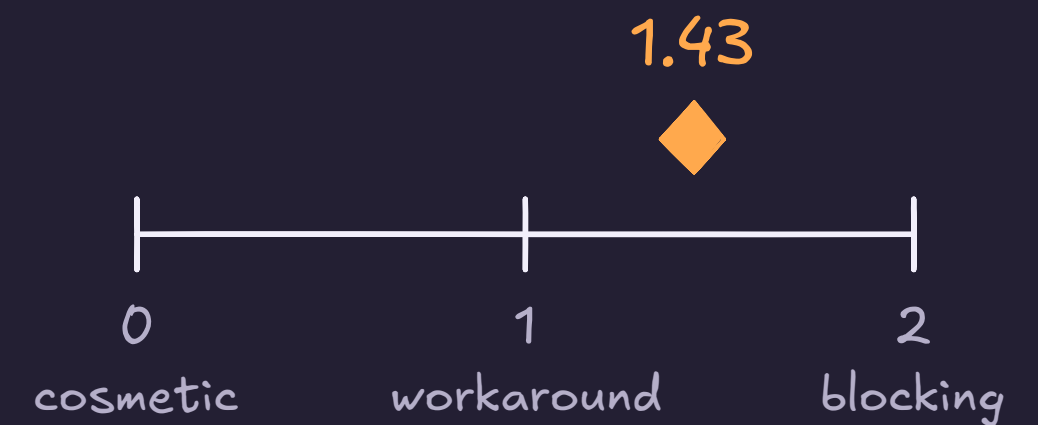
Choice

- ☒ billing
- ☐ sales
- ☐ support
- ☐ other

Which team? billing

a number on a scale

Score



How severe is this bug? 1.43

Every answer carries its probability, and Jev never answers outside your options.

Jev answers, a rule decides.

```
"outdated": {  
  "type": "noul",  
  "instructions": "Does one of the two memories  
    show that a claim in the other is outdated  
    or no longer true? ..."  
}
```

```
"the venue fixed the WiFi"  
  vs "the WiFi is unreliable"  
outdated           0.94  
new_reports_change 0.95  
same_question      0.79  
partial            0.20  
hypothetical       0.04
```

```
boolean conflict =  
    (sameQuestion >= 0.70 && differentAnswer >= 0.85)  
    || outdated >= 0.80  
    || changeReported >= 0.80;  
  
if (!conflict)           outcome = "none";  
else if (hypothetical >= 0.70) outcome = "review";  
else if (partial >= 0.70)   outcome = "review";  
else                       outcome = "superseded";
```

The thresholds live in code, and every answer is logged next to the decision.

It can be wrong, so it only votes.

TYPED ISN'T TRUE

Jev always answers in the format you asked for, including when it's wrong.

TWO GUARDS

In our tests it retired no memory that was still true once two guards were in: one for plans ("we might..."), one for exceptions ("except the MCP part").

STEERING

A planted sentence can push its answer. So the rule only touches older memories on the same subject, and every decision is logged.

A human can check every decision it made.

Great. Now make it
infrastructure.

Yes, that's HCL. Yes, that's Pulumi.

```
# Pulumi.yaml (trimmed)
name: agent-memory-infra
runtime: hcl
```

```
resource "ec_elasticsearch_project" "memory" {
  name          = var.project_name
  region_id     = var.region
  optimized_for = "general_purpose"

  metadata = {
    tags = {
      purpose = "agent-memory"
      managed = "pulumi"
    }
  }
}
```

```
provider "elasticstack" {
  elasticsearch {
    endpoints = [ec_elasticsearch_project.memory.endpoints.elasticsearch]
    username  = ec_elasticsearch_project.memory.credentials.username
    password  = ec_elasticsearch_project.memory.credentials.password
  }
  kibana {
    endpoints = [ec_elasticsearch_project.memory.endpoints.kibana]
    username  = ec_elasticsearch_project.memory.credentials.username
    password  = ec_elasticsearch_project.memory.credentials.password
  }
}
```



pulumi.com

Eight indices, two scoped keys.

```
resource "elasticsearch_index" "memory" {
  for_each = local.indices
  name     = each.key
  mappings = jsonencode({ properties = each.value })
}

resource "elasticsearch_security_api_key" "bridge" { # hooks + CLI
  role_descriptors = jsonencode({ agent_memory = {
    indices = [{ names = keys(local.indices), privileges = ["all"] }]
  } })
}

resource "elasticsearch_security_api_key" "mcp" { # any MCP client
  role_descriptors = jsonencode({ agent_memory_mcp = {
    indices = [{ names = keys(local.indices), privileges = ["read", "view_index_metadata"] }]
  } })
}
```



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One more provider, for what elasticstack can't create.

```
# Workflows only call out through Kibana's .http connector
resource "restapi_object" "jev_connector" {
  count = local.curation_enabled ? 1 : 0
  path  = "/api/actions/connector"
  data  = jsonencode(merge(local.jev_connector, { connector_type_id = ".http" }))

  ignore_changes_to      = ["secrets"]    # Kibana never returns them
  ignore_server_additions = true
}

resource "elasticstack_kibana_agentbuilder_workflow" "curation" {
  count = local.curation_enabled ? 1 : 0
  configuration_yaml = replace(file("workflows/memory-curation.yaml"),
    "__JEV_CONNECTOR_ID__", restapi_object.jev_connector[0].id)
}
```



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Demo.

Memory is infrastructure.
Treat it like infrastructure.

Q&A

Thanks.

Resources



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Slides + Demo

github.com/dirien/agent-memory

